

1 AssociationExplorer: A user-friendly Shiny application 2 for exploring associations and visual patterns

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10 Abstract

AssociationExplorer is an open-source interactive R Shiny application designed to help non-technical users explore statistical associations within multivariate datasets. Aimed particularly at journalists, educators, and engaged citizens, the tool facilitates the discovery and interpretation of meaningful patterns between variables without requiring programming or statistical expertise. Users can upload structured data (e.g., from surveys or open government datasets), select relevant variables, and dynamically visualize relationships via a correlation network and contextual bivariate plots. To illustrate its capabilities, we present a case study based on the European Social Survey (ESS), showcasing how users can investigate links between attitudes, behaviors, and socio-demographic indicators across countries. The app supports a range of association measures adapted to variable types (Pearson's r , Eta, and Cramer's V), ensuring both flexibility and statistical rigor. The visual interface enables users to adjust thresholds for association strength and examine results through interactive graphs and summary tables, making the app particularly well-suited for data storytelling, exploratory research, and public communication. AssociationExplorer demonstrates how open-source statistical tooling can enhance transparency, accessibility, and insight in the interpretation of complex social data.

11 **Keywords:** R Shiny, Exploratory data analysis, Correlation network

12 Metadata

13 The metadata associated with the current version of the software is summa-
14 rized in Table 1. The application has been developed and tested using R

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version 4.4.1 [1], ensuring compatibility with the package versions listed in the repository documentation.

Nr.	Code metadata description	Metadata
C1	Current code version	v3.5.7
C2	Permanent link to code/repository used for this code version	https://github.com/AntoineSoetewey/AssociationExplorer
C3	Permanent link to Reproducible Capsule	https://codeocean.com/capsule/7707308/tree
C4	Legal Code License	MIT License
C5	Code versioning system used	Git/GitHub
C6	Software code languages, tools, and services used	R, R Shiny
C7	Compilation requirements, operating environments & dependencies	See a list of the required R packages at https://github.com/AntoineSoetewey/AssociationExplorer/blob/main/packages.md
C8	If available link to developer documentation/manual/screencast	https://github.com/AntoineSoetewey/AssociationExplorer/tree/main/documentation
C9	Support for questions or issues	https://github.com/AntoineSoetewey/AssociationExplorer/issues

Table 1: Code metadata

The remainder of this paper is structured as follows. Section 1 outlines the motivation and significance of the application, highlighting its contribution to data accessibility and exploratory analysis. Section 2 provides a detailed description of the software’s architecture and functionalities. Section 3 presents an illustrative example using data from the European Social Survey. Section 4 discusses the potential impact of the application in research, education, and journalism. Finally, Section 5 concludes with a summary and future directions.

1. Motivation and significance

The growing availability of large, complex, and high-dimensional datasets in the social sciences and public policy domains offers unprecedented opportu-

28 nities for insight but also presents significant challenges for exploration and
 29 interpretation, particularly for non-specialist audiences. Journalists, educa-
 30 tors, and engaged citizens often struggle to identify and interpret meaningful
 31 relationships between variables without the aid of programming skills or for-
 32 mal statistical training. This barrier limits the broader societal impact of
 33 open data initiatives, which are designed to promote transparency, account-
 34 ability, and informed public discourse.

35 To address this gap, we developed AssociationExplorer, a free, open-source
 36 R Shiny [2] application that enables intuitive and statistically grounded ex-
 37 ploration of multivariate associations. The tool guides users through a visual
 38 journey of variable relationships by automatically computing appropriate bi-
 39 variate association measures—Pearson’s r , Eta, and Cramer’s V —depending
 40 on variable types, and presenting the results in an interactive correlation
 41 network. Users can set thresholds for the strength of association and explore
 42 linked bivariate plots or tables with descriptive labels. This workflow sup-
 43 ports transparent, reproducible, and non-technical exploratory data analysis
 44 (EDA).

45 Our software is particularly suited to survey-based datasets and public opin-
 46 ion studies. As an illustrative case, we apply AssociationExplorer to the
 47 European Social Survey (ESS), a cross-national survey that collects attitudi-
 48 nal, behavioral, and socio-demographic data across European countries. The
 49 tool allows users to uncover associations between trust in institutions, pol-
 50 icy preferences, media usage, and demographic characteristics without any
 51 coding. This type of interactive analysis can empower journalists to build
 52 data-driven narratives, educators to teach statistical thinking, and citizens
 53 to explore evidence underlying public debate.

54 While several tools and libraries exist for correlation analysis (e.g., `corrr`
 55 [3], `GGally` [4], `corrplot` [5], `ggstatsplot` [6], `correlation` [7, 8], `lares` [9]
 56 and `Hmisc` [10] in R [1], or Python packages like `seaborn` [11] and `pingouin`
 57 [12]), they typically require programming proficiency and focus primarily on
 58 numerical associations. Most of these tools do not handle nominal categorical
 59 variables directly; if included, such variables are often transformed using one-
 60 hot or dummy encoding, which can transform their original structure and
 61 limit interpretations.

62 In contrast, AssociationExplorer is designed to handle both quantitative and
 63 qualitative variables (including nominal factors) natively and transparently.
 64 It provides a guided, end-to-end workflow that begins with data upload and
 65 preprocessing, continues through variable selection and association filtering,
 66 and ends with interpretable visualizations. This structured process is in-
 67 tuitive and accessible for users of all backgrounds, making the app espe-
 68 cially suitable for those without programming experience or formal statistical

69 training. By lowering the technical barrier for statistical exploration, Asso-
70 ciationExplorer contributes to a more inclusive data culture and supports
71 data-driven discovery in both academic and public-facing contexts.

72 2. Software description

73 2.1. Software architecture

74 Upon upload, the dataset is preprocessed to exclude variables with zero vari-
75 ance, as these variables do not vary across observations and therefore cannot
76 contribute to meaningful associations or visualizations. Removing them helps
77 reduce noise and ensures that only informative variables are included in the
78 analysis. Optionally, the user can provide a variable description file, which
79 is integrated and used to annotate visual elements. The backend computes
80 association measures tailored to the variable types: Pearson's r for numeric
81 pairs, Cramer's V for categorical pairs, and the correlation ratio (η) for mixed
82 pairs. Associations are filtered using user-defined thresholds and represented
83 in a correlation network and complementary bivariate plots. The app handles
84 both CSV and Excel files.

85 The application is currently available as a standalone open-source R Shiny
86 application on GitHub.¹ It can be launched directly from an R session using
87 the following command, provided the `shiny` package is installed:

```
88 library(shiny)  
89 runGitHub("AssociationExplorer", "AntoineSoetewey")
```

90 A fully web-based version, requiring no installation or technical setup, is also
91 planned as part of its integration into a broader online platform developed
92 within the ODALON (Open multimodal Data for Automated Local News)
93 research project, which aims to support the production of local news (more
94 information about this project in Sections 3 and 4).

95 2.2. Software functionalities

96 The major functionalities of the AssociationExplorer application include:

- 97 • **Data upload and cleaning:** The app supports CSV and Excel files.
98 It automatically removes variables with only one unique value, as they
99 lack variability and cannot contribute to association analyses. Addi-
100 tionally, it can optionally integrate user-supplied descriptions of vari-
101 ables, which are used to enhance the clarity and interpretability of

¹<https://github.com/AntoineSoetewey/AssociationExplorer>

visualizations, particularly for non-technical users. The description file must be structured with two columns: **Variable** (containing the exact variable names as in the dataset) and **Description** (containing the corresponding descriptions).

- **Variable selection interface:** Users can interactively choose which variables to explore. When a description file is provided, a summary table links variable names to their descriptions.
- **Dynamic association filtering:** The app computes pairwise association measures between all selected variables, using a method tailored to the types of variables involved:

- For pairs of numeric variables X and Y , the app calculates Pearson’s correlation coefficient (r), and retains the association if the coefficient of determination (R^2) exceeds a user-defined threshold:

$$R^2 = r^2 = (\text{cor}(X, Y))^2 \quad (1)$$

where the Pearson’s correlation coefficient $\text{cor}(X, Y)$ is defined as:

$$r(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (2)$$

where $\bar{X}$ and $\bar{Y}$ are the sample means of X and Y , respectively, and n is the number of observations.

- For pairs of categorical variables, it computes Cramer’s V , a normalized measure of association derived from the chi-squared statistic:

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(k - 1, r - 1)}} \quad (3)$$

where χ^2 is the chi-squared statistic, n is the total number of observations, and k, r are the numbers of categories in each variable.

- For mixed pairs (one numeric and one categorical variable), the app computes the correlation ratio (η), which quantifies how much of the variance in the numeric variable is explained by the grouping structure of the categorical variable. It is defined as:

$$\eta = \sqrt{\frac{SS_{\text{between}}}{SS_{\text{total}}}} \quad (4)$$

where:

- SS_{total} is the *total sum of squares* of the numeric variable:

$$SS_{\text{total}} = \sum_{i=1}^n (y_i - \bar{y})^2$$

with y_i the observed numeric values and $\bar{y}$ their overall mean.

- SS_{between} is the *between-group sum of squares*, computed as:

$$SS_{\text{between}} = \sum_{g=1}^G n_g (\bar{y}_g - \bar{y})^2$$

where G is the number of groups (categories), n_g is the number of observations in group g , $\bar{y}_g$ is the group mean, and $\bar{y}$ is the overall mean.

This formulation captures the proportion of the total variance in the numeric variable that can be attributed to differences between the categorical groups. A pair is retained only if η^2 exceeds the numeric threshold defined by the user.

Each association is retained only if its corresponding strength metric— R^2 , η^2 , or Cramer’s V —exceeds the threshold set by the user. These thresholds can be adjusted interactively through the interface, and the filtering process is reactive: updates to the thresholds immediately propagate to the network and bivariate visualizations. This allows users to dynamically control the sensitivity of the association analysis and focus on relationships of substantive interest.

- **Interactive correlation network:** The filtered associations are displayed as an interactive graph where nodes represent variables and edges represent associations. Edge thickness and length reflect the strength of the association: stronger associations are shown with thicker and shorter edges, whereas weaker associations are displayed with thinner and longer edges. For quantitative–quantitative pairs specifically, the color of the edges conveys direction: red edges indicate negative associations, while blue edges indicate positive ones. This combination of visual cues helps users quickly identify the most meaningful relationships in the network. When hovering over nodes, variable descriptions are displayed if a description file has been provided by the user; otherwise, the variable names are shown by default. This allows for

157 quick access to additional context without cluttering the visualization,
158 enhancing interpretability while keeping the network clean and read-
159 able. The network is built using the `visNetwork` R package, which sup-
160 ports interactive and customizable graph layouts. Full documentation
161 is available at <https://datastorm-open.github.io/visNetwork/>.

162 • **Bivariate visualization of variable pairs:** For each variable pair
163 exceeding the threshold:

- 164 – Scatter plots with linear regression lines are shown for numeric
165 pairs, helping visualize the direction and strength of the relation-
166 ship. The regression line is computed using the ordinary least
167 squares (OLS) method, which minimizes the sum of squared ver-
168 tical distances between the observed data points and the fitted
169 line.
- 170 – Colored contingency tables with marginal sums are shown for cat-
171 egorical pairs, where cell background colors vary in intensity ac-
172 cording to the frequency of observations, using a blue gradient to
173 highlight higher counts.
- 174 – Mean plots are shown for numeric-categorical pairs, with bars or-
175 dered by mean value to make it easy to compare and rank cate-
176 gories based on the quantitative variable.

177 Confidence intervals for the regression lines and standard errors in the
178 mean plots are intentionally omitted to maintain a clean, uncluttered
179 visualization that prioritizes ease of interpretation. Mean plots were
180 selected over boxplots to avoid overwhelming non-expert users with
181 distributional information, focusing instead on clear, accessible insights
182 about average group differences.

183 • **Accessibility and user guidance:** A dedicated help section explains
184 each step, allowing users with a limited statistical background to inter-
185 actively explore their data.

186 2.3. Sample code snippets analysis

187 A representative excerpt from the application is provided in Code 1. This
188 snippet illustrates how the software selects the appropriate association mea-
189 sure based on the types of the variable pair and applies user-defined thresh-
190 olds to retain only sufficiently strong associations.

191

```

# Numeric vs numeric case
if (is_num1 && is_num2) {
  ...
  r <- cor(x, y, use = "complete.obs")
  cor_val <- ifelse(r^2 >= threshold_num, r, 0)
  cor_type <- "Pearson"

# Categorical vs categorical case
} else if (!is_num1 && !is_num2) {
  ...
  tbl <- table(x, y)
  ...
  n_obs <- sum(tbl)
  df_min <- min(nrow(tbl) - 1, ncol(tbl) - 1)
  if (df_min > 0) {
    v_cramer <- sqrt(chi$statistic / (n_obs * df_min))
    cor_val <- ifelse(v_cramer >= threshold_cat,
                      v_cramer, 0)
    cor_type <- "Cramer"
  }

# Mixed case (numeric vs categorical)
} else {
  ...
  means_by_group <- tapply(num_var, cat_var,
                           mean, na.rm = TRUE)
  overall_mean <- mean(num_var, na.rm = TRUE)
  n_groups <- tapply(num_var, cat_var, length)
  bss <- sum(n_groups * (means_by_group - overall_mean)^2,
            na.rm = TRUE)
  tss <- sum((num_var - overall_mean)^2, na.rm = TRUE)

  if (tss > 0) {
    eta <- sqrt(bss / tss)
    cor_val <- ifelse(eta^2 >= threshold_num, eta, 0)
    cor_type <- "Eta"
  }
}

```

Code 1: Selection of the association measure according to variable types.

192 This conditional structure ensures that the correct statistical method is ap-
193 plied for each type of variable pair, supporting a robust and interpretable
194 exploration of associations.

195 3. Illustrative example

196 To demonstrate the core functionalities of AssociationExplorer, we use a cu-
197 rated subset of data from the European Social Survey (ESS), Round 11.
198 The ESS is a large-scale, cross-national survey that measures attitudes, be-
199 liefs, and behaviors across European countries. The original dataset includes
200 responses from over 46,000 individuals on topics such as politics, trust, well-
201 being, media use, and health. The full ESS dataset, codebook, and docu-
202 mentation are freely available at <https://ess.sikt.no/en/> [13, 14].

203 For this example, we focus on the Belgian respondents, resulting in a re-
204 duced dataset of 1,594 individuals. We selected 60 variables covering areas
205 highly relevant for understanding public opinion and everyday life in Belgium:
206 interest in politics, confidence in institutions, lifestyle behaviors, perceived
207 discrimination, vaccination, and more. These variables include both numbers
208 (quantitative data) and labels or categories (qualitative data), making the
209 dataset ideal for exploring diverse forms of associations.

210 This example is particularly relevant for our research project ODALON,
211 which aims to develop a platform that supports the (semi-)automated pro-
212 duction of local news in Belgium. AssociationExplorer plays a key role in
213 this effort by offering journalists, researchers, and citizens an intuitive tool to
214 explore potentially newsworthy patterns in public and survey data, without
215 requiring programming skills or statistical training.

216 Data preparation for the curated dataset was carried out in R and included:

- 217 • Filtering the dataset to include only Belgian respondents.
- 218 • Converting survey-specific nonresponse codes (e.g., 77, 88, 9999, etc.)
219 to NA values, based on the ESS codebook.
- 220 • Recoding several categorical variables to have meaningful and inter-
221 pretable labels (e.g., for gender, religion, political participation, or
222 health behaviors).

223 The full R script used to perform this transformation is openly available in the
224 data folder of the GitHub repository at [https://github.com/AntoineSoetewey/](https://github.com/AntoineSoetewey/AssociationExplorer/tree/main/data)
225 `AssociationExplorer/tree/main/data`.

226 Once the dataset is uploaded into AssociationExplorer via the Data tab (see
227 Figure 1), users are guided through a step-by-step process. In addition to

Association Explorer

Upload your dataset (CSV or Excel)

Browse... ESS11_BE_data.csv
Upload complete

(Optional) Upload variable descriptions (CSV or Excel)

Browse... ESS11_BE_description.csv
Upload complete

The descriptions file must contain exactly two columns named 'Variable' and 'Description'.

Process data

Figure 1: Data upload tab. Users upload their dataset and can optionally provide a variable description file containing variable names and descriptions.

the main dataset, users can optionally upload a separate description file that provides human-readable explanations for each variable. In our example, this file was created using information from the official ESS codebook, allowing for clearer interpretation throughout the app interface. If no description file is provided, the application will automatically use the variable names themselves as default labels in all visualizations and summary tables.

In the **Variables** tab, they select the variables they wish to explore, optionally assisted by a table of the variables' names and descriptions (see Figure 2).

Next, users can adjust the association thresholds in the **Correlation Network** tab to focus on the most meaningful relationships among the selected variables (see Figure 3).

Finally, the **Pairs Plots** tab displays detailed bivariate visualizations, including scatter plots, mean plots, and contingency tables, for each retained association (see Figures 4-6).

In addition, the **Help** tab provides a concise, step-by-step guide on how to use the application (see Figure 7).

Each visualization panel includes contextual download controls to facilitate the reuse of figures and tables (e.g., in reports or data-journalism pieces). The correlation network can be exported directly as a PNG image, and each pairwise association displayed in the Pairs Plots tab can likewise be downloaded as an image. Scatterplots, mean plots, and contingency tables are saved with the same styling and labels as shown on screen, ensuring full vi-

Association Explorer

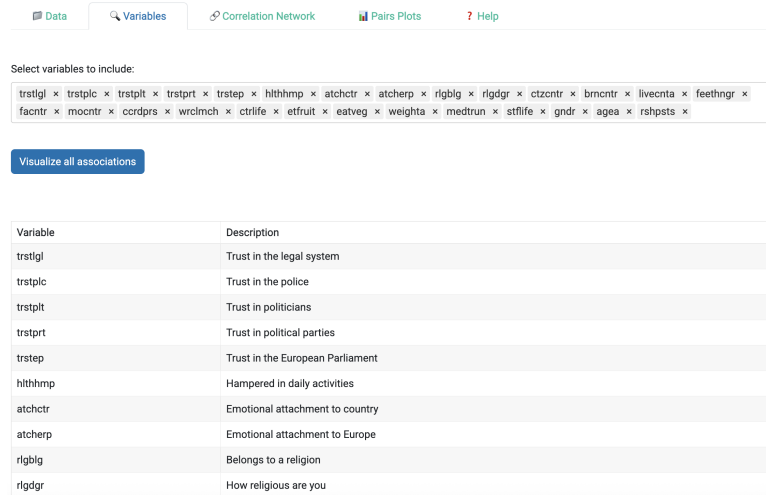


Figure 2: Variables tab. Users select the variables to include in the analysis, with access to descriptions (if provided in the data tab) for easier selection.

Association Explorer

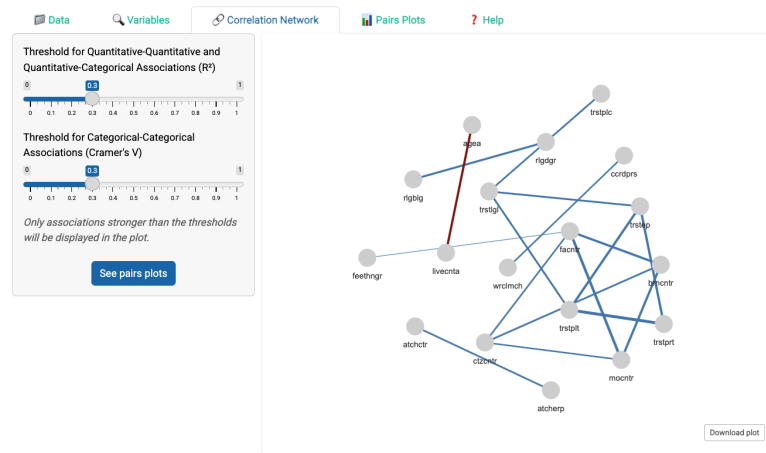


Figure 3: Interactive correlation network. Nodes represent variables and edges represent associations; edge thickness and length indicate strength, and edge color reflects direction (for numeric pairs only).

Association Explorer

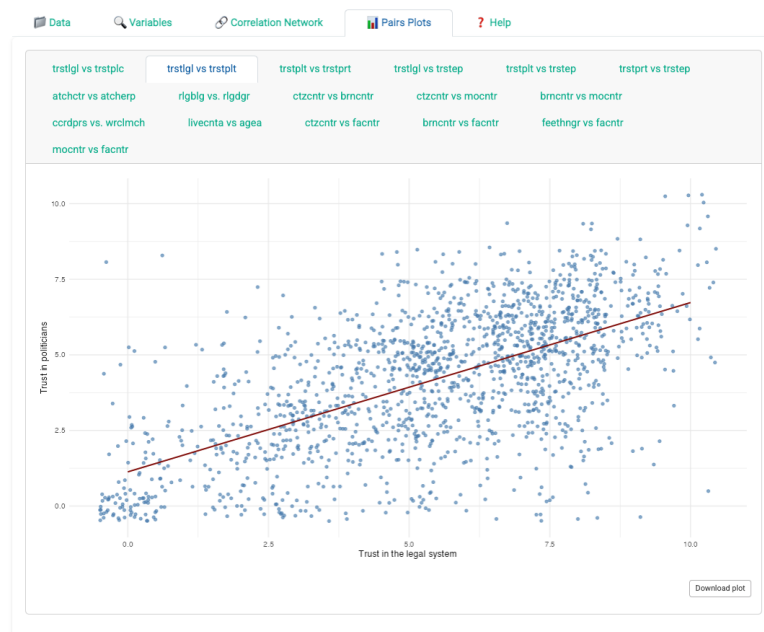


Figure 4: Example of a scatter plot for a numeric-numeric pair. The plot includes a linear regression line fitted via ordinary least squares (OLS).

Association Explorer

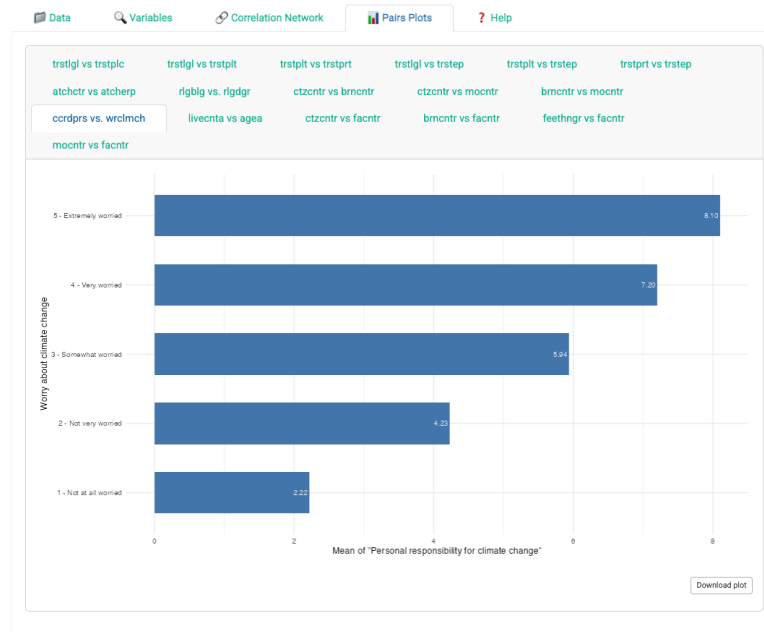


Figure 5: Example of a mean plot for a numeric-categorical pair. Categories are ordered by the mean value of the quantitative variable to aid interpretation.

Association Explorer

The figure is a contingency table titled 'Born in country' on the columns and 'Citizen of country' on the rows. The cells are colored based on frequency, with the highest frequency cell (Yes/Yes) in dark blue. Marginal sums are shown in the last column. A 'Download table' button is at the bottom right.

Citizen of country	Born in country		Sum
	No	Yes	
No	102	19	121
Yes	112	1,360	1,472
Sum	214	1,379	1,593

Figure 6: Example of a colored contingency table for a categorical-categorical pair, with color intensity reflecting frequencies and marginal sums displayed.

Association Explorer

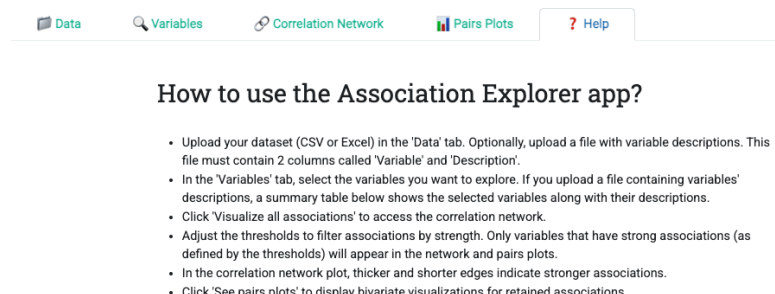


Figure 7: Help tab, offering guidance on how to use the application.

sual consistency between the interactive interface and the exported outputs. This example shows how non-expert users, such as journalists or engaged citizens, can uncover unexpected or important relationships in a public opinion dataset. These insights can serve as the starting point for local news, public debates, or policy communication. A screencast demonstrating the full workflow of the application is available in the documentation folder of the GitHub repository.

Note that in this example, no survey weights have been applied. While the ESS dataset includes weights to account for complex sampling designs, we chose not to use them here in order to maintain clarity and responsiveness in the interface, and because our focus is on demonstrating the app's core functionality rather than producing nationally representative statistics.

4. Impact

AssociationExplorer enables a broader range of users, including researchers, students, journalists, and engaged citizens, to explore complex datasets involving both quantitative and qualitative variables. By eliminating the need for advanced programming or statistical knowledge, the software lowers the barrier to entry for data-driven discovery and interpretation. The app facilitates the systematic screening of pairwise associations in survey or public data, offering insights that can guide hypothesis generation, variable selection, or further multivariate modeling.

It also facilitates the assessment of relationships between variables of different types (numeric, ordinal, and nominal) within a single unified interface. Unlike many tools that require preprocessing steps such as dummy coding, AssociationExplorer handles categorical variables natively and automatically selects the appropriate statistical association measure (i.e., Pear-

277 son's r , Cramer's V , correlation ratio η). This ensures that users can obtain
 278 meaningful and interpretable results without needing to master complex pre-
 279 processing steps or statistical diagnostics.
 280 Beyond supporting research, we are convinced that automating these statisti-
 281 cal processes can contribute to the promotion of open public data repositories.
 282 In fact, the scientific literature points to a chronic underuse of open public
 283 data [15], particularly due to the resources and knowledge required to pro-
 284 cess these data. By lowering this barrier to entry, this tool makes it possible
 285 to utilize databases that would otherwise remain inactive. The exploitation
 286 and understanding of these data responds to crucial issues of transparency
 287 in public action [16].
 288 Although this tool is open to anyone, its development was particularly in-
 289 spired by its use in a journalistic context. Data journalism, in particular, is
 290 one of those examples of virtuous practice that we believe is best suited to
 291 supporting the open data movement. This practice, which relies on the use
 292 of digital data as material for journalistic investigation [17], is now used to
 293 analyze election results, explain climate change, and track the evolution of
 294 health crises. However, it comes at a cost, which explains why this practice
 295 is still limited to a few large national media outlets.
 296 However, it would be detrimental to leave these practices inaccessible to local
 297 newsrooms. By covering events that are often overlooked by national media,
 298 these local media outlets can highlight data that would otherwise be ignored
 299 (finance, consumption, police, municipal council records, etc.). Supporting
 300 data journalism at the local level can also help to enhance the value of this
 301 journalistic practice, which has been shown to strengthen social cohesion [18],
 302 stimulate civic participation [19], and influence public policies [20].
 303 While national media outlets are struggling, these challenges are even more
 304 pronounced for local media, which generally face the same problems but
 305 with even more limited resources [21]. As a result, few institutions have the
 306 means to meet the labor and skill requirements necessary for this investigative
 307 practice [22]. We hope that AssociationExplorer can help meet these needs.
 308 The app is also planned to be used as an interactive teaching aid in statistics
 309 and data literacy courses. It will allow students to upload real datasets and
 310 immediately visualize how variables relate, reinforcing theoretical concepts
 311 through intuitive, hands-on exploration. By presenting associations visually,
 312 the app encourages learners to engage with their data more actively and
 313 reflect critically on patterns, variable relationships, and data quality.
 314 Although already fully functional, the software is currently made available as
 315 a standalone open-source R Shiny application on GitHub and can be launched
 316 directly from R. At this stage, it is therefore only accessible to users who have
 317 R installed on their machine. As part of a broader research project called

318 ODALON, which aims to build an integrated platform combining multiple
319 tools for automated local news production, AssociationExplorer will serve
320 as a key building block. A public, fully online version of the app will be
321 released as the project progresses, and no later than December 2026 (which
322 corresponds to the end of the project), ensuring that anyone, including those
323 without access to R, can freely use the application through a user-friendly
324 web interface.

325 Future impact is expected through integration into platforms for civic data
326 storytelling, local journalism, and public communication. While not intended
327 for commercialization, the application’s role in supporting open, inclusive,
328 and responsible data exploration reflects its potential for broader societal
329 relevance.

330 5. Conclusions

331 AssociationExplorer provides a user-friendly, open-source solution for explor-
332 ing statistical associations within multivariate datasets. By integrating ro-
333 bust statistical measures with dynamic and intuitive visualizations, the app
334 empowers non-expert users, such as journalists, educators, and engaged cit-
335 izens, to uncover patterns and relationships that might otherwise remain
336 hidden. Its ability to handle both quantitative and qualitative variables,
337 combined with an interactive interface and an easy-to-follow workflow, makes
338 it especially suitable for exploratory research, data storytelling, and public
339 communication.

340 Beyond its technical features, the application contributes to democratizing
341 access to statistical tools and supports data literacy in a wide range of con-
342 texts. The case study using the European Social Survey illustrates how users
343 can navigate from raw data to meaningful insights with minimal technical
344 knowledge.

345 While the app is currently accessible through R for users with basic techni-
346 cal setup, a fully web-based version will make it universally available as part
347 of a broader data exploration platform being developed under the ODALON
348 project. Before public deployment, performance will be systematically tested
349 on larger datasets to ensure both responsiveness and reliability. Any iden-
350 tified limitations, such as reduced interactivity with high-dimensional data
351 or memory constraints, will be clearly communicated to users through the
352 application’s documentation. By lowering barriers to data exploration and in-
353 terpretation, AssociationExplorer supports more inclusive, transparent, and
354 evidence-informed engagement with complex social data.

355 Future improvements will be guided by user feedback and community con-
356 tributions. One planned feature is the ability to incorporate user-specified

weights, especially relevant for survey data. However, this functionality is not yet implemented in the current version to preserve simplicity and ensure responsiveness in the user interface, particularly for users unfamiliar with survey weighting procedures. In the same spirit, future versions will also include a dedicated panel for basic graphic customization (e.g., color palettes, titles, and labels). Although not implemented at the current stage to preserve a lightweight and responsive interface for non-technical users, this feature is planned for the forthcoming public web-based release to enhance usability and flexibility.

Additional avenues for future research and development include extending the application to compute partial correlations, which can help isolate direct relationships between variables by controlling for confounders. Visualizations involving three variables could also be introduced, for instance, scatter plots where the size or color of points reflects a third variable, to uncover more nuanced patterns. Furthermore, users could be allowed to filter variables based on missingness thresholds or variable types, and a module for longitudinal data (e.g., panel structures or time series) could expand the app’s relevance to temporal analyses. For contingency tables, adding conditional distributions (by row or by column) in addition to the joint distribution could provide more informative insights. This would allow users to better explore subsets of the data. For example, even if a group such is underrepresented in the sample (and therefore lightly colored in the joint table), the conditional distribution would still reveal the relative patterns within that group. Another enhancement could involve enabling two-sided intervals for the association thresholds, rather than the current one-sided sliders. This would allow users not only to focus on the strongest associations, but also to deliberately examine the weakest ones, thereby uncovering cases where variables appear largely independent. In addition, users could be given the option to flip the X and Y axes in quantitative–quantitative pairs plots, providing greater flexibility when comparing variables and improving interpretability when relationships are more intuitively viewed in the reverse orientation. Providing interpretive text alongside tables and plots (especially if the application is deployed for non-specialist audiences) could also help ensure that outputs are understood correctly and used responsibly. Lastly, although the application includes built-in export options for visualizations and tables, future extensions could incorporate a dedicated panel for basic graphic customization, and even automated summaries of key associations, to further enhance its use in reporting and collaborative workflows.

A more ambitious extension would integrate a large language model (LLM) to enable users to enter a free-text prompt or a thematic query. The app would then automatically identify variables that are semantically similar or

potentially associated with the theme, based on the dataset and the optional description file provided by the user. These extracted variables would be pre-selected in the interface, as if the user had chosen them manually, after which the analysis would proceed as in the current version. This feature could significantly enhance accessibility for users unfamiliar with the dataset or its structure, and accelerate the discovery of relevant associations.

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